



US 20250275594A1

(19) **United States**

(12) **Patent Application Publication**
BARBEAU et al.

(10) **Pub. No.: US 2025/0275594 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **LOW-CARBON FOOTPRINT OUTER
SHELLS FOR FIREFIGHTER GARMENTS**

Publication Classification

(51) **Int. Cl.**

A41D 27/04 (2006.01)

A41D 13/00 (2006.01)

(52) **U.S. Cl.**

CPC *A41D 27/04* (2013.01); *A41D 13/0002*
(2013.01)

(71) Applicant: **Innotex Inc.**, Richmond (CA)

(72) Inventors: **Claude BARBEAU**, Racine (CA);
Karine MARTIN, Magog (CA)

(21) Appl. No.: **19/066,467**

(22) Filed: **Feb. 28, 2025**

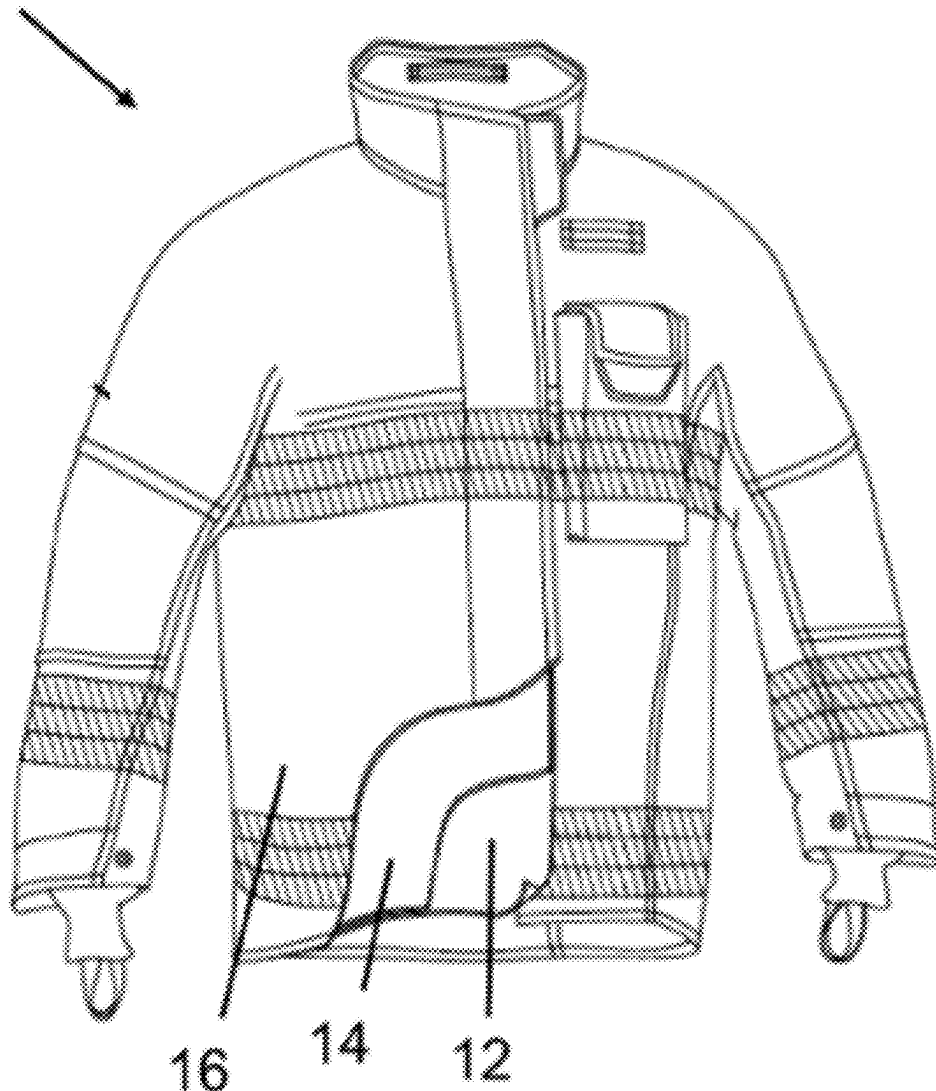
Related U.S. Application Data

(60) Provisional application No. 63/560,075, filed on Mar.
1, 2024.

ABSTRACT

There is provided a garment, including a first liner defining an innermost layer of the firefighter protective garment; a second liner extending over the first liner; an outer shell defining an outermost layer of the firefighter protective garment and extending over the second liner, the outer shell having: a first surface made from a first material; and a second surface opposite the first surface, the second surface being made from a second material. At least one or only one of the first material and the second material is at least partially composed of recycled fibers.

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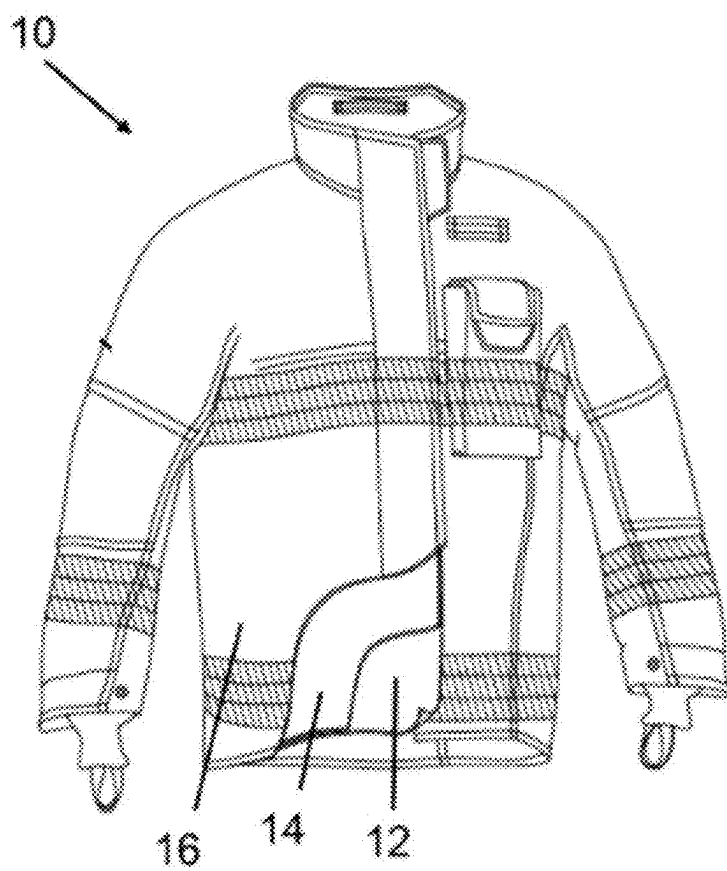


FIGURE 1

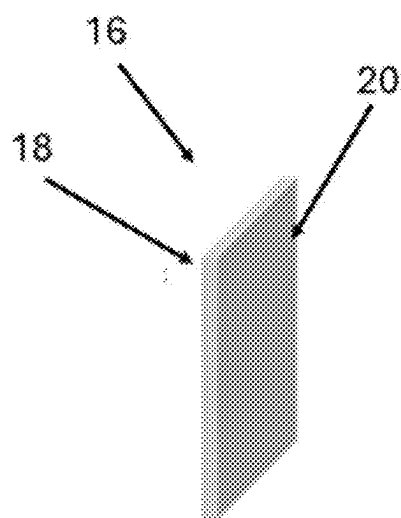


FIGURE 2

LOW-CARBON FOOTPRINT OUTER SHELLS FOR FIREFIGHTER GARMENTS

TECHNICAL FIELD

[0001] The technical field generally relates to an outer shell for protective garments and more particularly concerns a firefighter protective garment including such an outer shell.

BACKGROUND

[0002] Firefighter protective garments, such as coats and pants, have designs which are required to be compliant with the National Fire Protection Association Standard on Protective Ensembles for Structural Fire Fighting and Proximity Fire Fighting.

[0003] There remains a need in the art for interfaces of firefighter garments that improve on the security and/or comfort of the wearer, as well as their potential environmental impacts, while being compliant with the actual standards.

SUMMARY

[0004] In accordance with one aspect, there is provided a firefighter protective garment. The firefighter protective garment includes a first liner defining an innermost layer of the firefighter protective garment; a second liner extending over the first liner; and an outer shell defining an outermost layer of the firefighter protective garment and extending over the second liner. The outer shell has a first surface made from a first material; and a second surface opposite the first surface, the second surface being made from a second material. At least one of the first material and the second material is at least partially composed of recycled fibers.

[0005] In some embodiments, the first liner is a thermal barrier.

[0006] In some embodiments, the second liner is a moisture barrier.

[0007] In some embodiments, the first material includes about 25% of recycled fibers.

[0008] In some embodiments, the first material includes Nomex® IIIA.

[0009] In some embodiments, the second material includes para-aramid fibers.

[0010] In some embodiments, the second material includes meta-aramid fibers.

[0011] In some embodiments, the second material includes polybenzimidazole (PBI).

[0012] In some embodiments, the second material includes at least 25% of recycled fibers.

[0013] In some embodiments, the firefighter protective garment includes a firefighter protective coat.

[0014] In some embodiments, the firefighter protective garment includes firefighter protective pants.

[0015] In accordance with one aspect, there is provided a firefighter protective garment. The firefighter protective garment includes a first liner defining an innermost layer of the firefighter protective garment; a second liner extending over the first liner; and an outer shell defining an outermost layer of the firefighter protective garment and extending over the second liner. The outer shell has a first surface made from a first material; and a second surface opposite the first surface, the second surface being made from a second material,

wherein only one of the first material and the second material is at least partially composed of recycled fibers.

[0016] In some embodiments, the first liner is a thermal barrier.

[0017] In some embodiments, the second liner is a moisture barrier.

[0018] In some embodiments, the second material includes recycled fibers.

[0019] In some embodiments, the first material includes multifilament fibers.

[0020] In some embodiments, the first material includes meta-aramid fibers.

[0021] In some embodiments, the second material includes para-aramid fibers.

[0022] In some embodiments, the second material includes meta-aramid fibers.

[0023] In some embodiments, the second material includes polybenzimidazole (PBI).

[0024] In some embodiments, the second material includes at least 25% of recycled fibers.

[0025] In some embodiments, the firefighter protective garment includes a firefighter protective coat.

[0026] In some embodiments, the firefighter protective garment includes firefighter protective pants.

[0027] In accordance with one aspect, there is provided an outer shell for a firefighter protective garment. The outer shell has a first surface made from a first material; and a second surface opposite the first surface, the second surface being made from a second material, wherein at least one of the first material and the second material is at least partially composed of recycled fibers.

[0028] In accordance with one aspect, there is provided an outer shell for a firefighter protective garment. The outer shell has a first surface made from a first material; and a second surface opposite the first surface, the second surface being made from a second material, wherein only one of the first material and the second material is at least partially composed of recycled fibers.

[0029] Other features and advantages of the present invention will be better understood upon a reading of embodiments thereof with reference to the appended drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1 illustrates a firefighter protective garment, in accordance with one embodiment.

[0031] FIG. 2 shows a portion of an outer shell for a firefighter protective garment, in accordance with one embodiment.

DETAILED DESCRIPTION

[0032] In the present description, similar features in the drawings have been given similar reference numerals. To avoid cluttering certain figures, some elements may not have been indicated if they were already identified in a preceding figure. It should also be understood that the elements of the drawings are not necessarily depicted to scale, since emphasis is placed on clearly illustrating the elements and structures of the present embodiments. Furthermore, positional descriptors indicating the location and/or orientation of one element with respect to another element are used herein for ease and clarity of description. Unless otherwise indicated, these positional descriptors should be taken in the context of the figures and should not be considered limiting. More

particularly, it will be understood that such spatially relative terms are intended to encompass different orientations in the use or operation of the present embodiments, in addition to the orientations exemplified in the figures.

[0033] The terms “a”, “an” and “one” are defined herein to mean “at least one”, that is, these terms do not exclude a plural number of items, unless stated otherwise.

[0034] Terms such as “substantially”, “generally” and “about”, that modify a value, condition or characteristic of a feature of an exemplary embodiment, should be understood to mean that the value, condition or characteristic is defined within tolerances that are acceptable for the proper operation of this exemplary embodiment for its intended application.

[0035] Unless stated otherwise, the terms “attached”, “connected” and “coupled”, and derivatives and variants thereof, refer herein to any structural or functional connection or coupling, either direct or indirect, between two or more elements. For example, the connection or coupling between the elements may be acoustical, mechanical, optical, electrical, thermal, logical, or any combinations thereof.

[0036] The present description refers to protective garments, such as firefighter protective coats and pants. Of note, passages describing the firefighter protective coats could also be applied to firefighter protective pants, and vice-versa.

[0037] In accordance with one aspect, and with reference to FIG. 1, there is provided a firefighter protective garment 10. In FIG. 1, the firefighter protective garment 10 is embodied by a firefighter protective coat. The firefighter protective garment 10 includes a first liner 12 defining an innermost layer of the firefighter protective garment 10, a second liner 14 extending over the first liner 12, and an outer shell 16 defining an outermost layer of the firefighter protective garment 10 and extending over the second liner 14.

[0038] As better illustrated in FIG. 2, the outer shell 16 has a first surface 18 and a second surface 20 opposite the first surface 18. The first surface 18 is made from a first material and the second surface 20 is made from a second material. At least one of the first material and the second material is at least partially composed of recycled fibers. In some embodiments, only the first material of the first surface 18 is at least partially composed of recycled fibers. In some embodiments, only the second material of the second surface 20 is at least partially composed of recycled fibers. In some embodiments, both the first material of the first surface 18 and the second material of the second surface 20 are at least partially composed of recycled fibers.

[0039] In some embodiments, the first liner 12 is a thermal barrier. In some embodiments, the second liner 14 is a moisture barrier. It should be noted that the layers forming the firefighter protective garment 10 may be disposed in different sequences, depending, for example and without being limitative, on the targeted application or requirements associated with the targeted application. Other layer(s) and/or liner(s) could also be added in the design of the firefighter protective garment 10.

[0040] Now that the components included in the firefighter protective garment 10 have been described, different embodiments of the first material, forming the first surface 18, and the second material, forming the second surface 20, will be presented

[0041] In some embodiments, the first material includes about 25% of recycled fibers. In some embodiments, the composition of the first material may include less than 25%

of recycled fibers. In some embodiments, the composition of the first material may include less than 25% of recycled fibers. In some embodiments, the first material includes Nomex® IIIA. In some embodiments, the first material may include a combination of the fibers listed above.

[0042] In some embodiments, the second material includes para-aramid fibers. In some embodiments, the second material includes meta-aramid fibers. In some embodiments, the second material includes polybenzimidazole (PBI). In some embodiments, the second material includes at least 25% of recycled fibers. In some embodiments, the composition of the second material may include less than 25% of recycled fibers. In some embodiments, the composition of the second material may include less than 25% of recycled fibers. In some embodiments, the second material may include a combination of the fibers listed above.

[0043] In some embodiments, the firefighter protective garment 10 includes a firefighter protective coat. In some embodiments, the firefighter protective garment includes firefighter protective pants.

[0044] In accordance with one aspect, there is provided a firefighter protective garment. The firefighter protective garment includes a first liner defining an innermost layer of the firefighter protective garment; a second liner extending over the first liner; and an outer shell defining an outermost layer of the firefighter protective garment and extending over the second liner. The outer shell has a first surface made from a first material; and a second surface opposite the first surface, the second surface being made from a second material, wherein only one of the first material and the second material is at least partially composed of recycled fibers. The firefighter protective garment according to this aspect is similar to the one previously described but differs in that that only one of the first and second materials contains recycled fibers. In some embodiments, the first material contains recycled fibers, and the second material does not contain recycled fibers. In some embodiments, the second material contains recycled fibers, and the first material does not contain recycled fibers.

[0045] In some embodiments, the first liner is a thermal barrier. In some embodiments, the second liner is a moisture barrier. It should be noted that the layers forming the firefighter protective garment may be disposed in different sequences, depending, for example and without being limitative, on the targeted application or requirements associated with the targeted application. Other layer(s) and/or liner(s) could also be added in the design of the firefighter protective garment.

[0046] In some embodiments, the second material includes recycled fibers. In some embodiments, the first material includes multifilament fibers. In some embodiments, the first material includes meta-aramid fibers. In some embodiments, the second material includes para-aramid fibers. In some embodiments, the second material includes meta-aramid fibers. In some embodiments, the second material includes polybenzimidazole (PBI). In some embodiments, the second material includes at least 25% of recycled fibers. In some embodiments, the first material and/or the second material may include a combination of the fibers listed above.

[0047] In some embodiments, the firefighter protective garment includes a firefighter protective coat. In some embodiments, the firefighter protective garment includes firefighter protective pants.

[0048] In accordance with one aspect, there is provided an outer shell for a firefighter protective garment. The outer shell has a first surface made from a first material; and a second surface opposite the first surface, the second surface being made from a second material, wherein at least one of the first material and the second material is at least partially composed of recycled fibers.

[0049] In accordance with one aspect, there is provided an outer shell for a firefighter protective garment. The outer shell has a first surface made from a first material;

[0050] and a second surface opposite the first surface, the second surface being made from a second material, wherein only one of the first material and the second material is at least partially composed of recycled fibers.

[0051] The following two tables illustrate examples of embodiments of the materials forming the outer shell, as well as corresponding compositions.

Specification											
	Type	①	Weight②	Color	①	②	③	Option 1 Fibers	Option 2 ① party	Type②	%
External fabric	2 faces	②	7.0	Piece dye	WR	A	Face	②, ③		Recycled	25%
							Back	③		Recycled	Min 25%
						B	Face		③		
Liner	Woven fabric						Back	③	③	Recycled	Min 25%
	Non-woven fabric										

Note②

Without②

② indicates text missing or illegible when filed

Specification											
	Type	①	Weight②	Color	①	②	③	Option 1 Fibers	Option 2 Dupont	Type②	%
External fabric	2 faces	②	7.0	Piece dye	WR	A	Face	②, ③		Recycled	25%
							Back	③/Mets	③	Recycled	Min 25%
						B	Face		③		
							Back	③	③	Recycled	Min 25%

② indicates text missing or illegible when filed

[0052] Several alternative embodiments and examples have been described and illustrated herein. The embodiments described above are intended to be exemplary only. A person skilled in the art would appreciate the features of the individual embodiments, and the possible combinations and variations of the components. A person skilled in the art would further appreciate that any of the embodiments could be provided in any combination with the other embodiments disclosed herein. The present examples and embodiments, therefore, are to be considered in all respects as illustrative and not restrictive. Accordingly, while specific embodiments have been illustrated and described, numerous modifications come to mind without significantly departing from the present disclosure and the appended claims.

1. A firefighter protective garment, comprising:

a first liner defining an innermost layer of the firefighter protective garment;

a second liner extending over the first liner; and

an outer shell defining an outermost layer of the firefighter protective garment and extending over the second liner, the outer shell having:

a first surface made from a first material; and

a second surface opposite the first surface, the second surface being made from a second material, wherein at least one of the first material and the second material is at least partially composed of recycled fibers.

2. The firefighter protective garment of claim 1, wherein the first liner is a thermal barrier.

3. The firefighter protective garment of claim 1, wherein the second liner is a moisture barrier.

4. The firefighter protective garment of claim 1, wherein the first material comprises about 25% of recycled fibers.

5. The firefighter protective garment of claim 1, wherein the first material comprises Nomex® IIIA.

6. The firefighter protective garment of claim 1, wherein the second material comprises para-aramid fibers.

7. The firefighter protective garment of claim 1, wherein the second material comprises meta-aramid fibers.

8. The firefighter protective garment of claim 1, wherein the second material comprises polybenzimidazole (PBI).

9. The firefighter protective garment of claim 1, wherein the second material comprises at least 25% of recycled fibers.

10. The firefighter protective garment of claim 1, wherein the firefighter protective garment comprises a firefighter protective coat.

11. The firefighter protective garment of claim 1, wherein the firefighter protective garment comprises firefighter protective pants.

- 12.** A firefighter protective garment, comprising:
a first liner defining an innermost layer of the firefighter protective garment;
a second liner extending over the first liner; and
an outer shell defining an outermost layer of the firefighter protective garment and extending over the second liner, the outer shell having:
a first surface made from a first material; and
a second surface opposite the first surface, the second surface being made from a second material, wherein only one of the first material and the second material is at least partially composed of recycled fibers.
- 13.** The firefighter protective garment of claim **12**, wherein the first liner is a thermal barrier.
- 14.** The firefighter protective garment of claim **12**, wherein the second liner is a moisture barrier.
- 15.** The firefighter protective garment of claim **12**, wherein the second material comprises recycled fibers.
- 16.** The firefighter protective garment of claim **12**, wherein the first material comprises multifilament fibers.
- 17.** The firefighter protective garment of claim **12**, wherein the first material comprises meta-aramid fibers.
- 18.** The firefighter protective garment of claim **12**, wherein the second material comprises para-aramid fibers.
- 19.** The firefighter protective garment of claim **12**, wherein the second material comprises meta-aramid fibers.

20. The firefighter protective garment of claim **12**, wherein the second material comprises polybenzimidazole (PBI).

21. The firefighter protective garment of claim **12**, wherein the second material comprises at least 25% of recycled fibers.

22. The firefighter protective garment of claim **12**, wherein the firefighter protective garment comprises a firefighter protective coat.

23. The firefighter protective garment of claim **12**, wherein the firefighter protective garment comprises firefighter protective pants.

24. An outer shell for a firefighter protective garment, the outer shell having:

- a first surface made from a first material; and
- a second surface opposite the first surface, the second surface being made from a second material, wherein at least one of the first material and the second material is at least partially composed of recycled fibers.

25. An outer shell for a firefighter protective garment, the outer shell having:

- a first surface made from a first material; and
- a second surface opposite the first surface, the second surface being made from a second material, wherein only one of the first material and the second material is at least partially composed of recycled fibers.

* * * * *